

F722 — S2, the make-or-break, run on the SOURCED type-IV embedding: the $5 \rightarrow 4$ projection drops EXACTLY the e-singlet direction, and O is the (2,2) bi-doublet — a DIFFERENT irrep — so by Schur orthogonality O carries no e-component and $v = O \neq 0$. S2 CLEARS (non-central), target-innocent. This is not “O is a non-singlet so $v \neq 0$ ” (the argument Casey correctly forbade — the dimension drop is where that fails); it is the sharper fact that the dropped dimension is a specific irrep (the (1,1) singlet = the Jordan identity e), O is a distinct irrep (the (2,2)), and distinct K-irreps are orthogonal under the trace form — so O literally cannot land in the dropped direction. The one residual is Elie’s numerical confirmation on the explicit embedding; the structure forces $v \neq 0$. Bonus: the count node reconciles to the rank (K975 watch-item 1) — interior idempotents = {muon $\ell=1$, electron $\ell=2$ }, boundary +1 = tau $\ell=0$, superseding K947’s {tau,muon}. Tier: S2 clears on sourced structure (FK Ch III + F603), pending Elie’s numeric + Cal’s “v non-central?” audit; if it holds, the muon is $S1 \otimes S2 \otimes S3 \otimes S4 \otimes \text{interior-2} \rightarrow \text{Keeper fires K967} \rightarrow \text{DERIVED}$. I do NOT bank it here — Keeper fires blind.

Lyra, Tue 2026-07-28. The one gate. Sourced, not reconstructed — the dropped dimension turns out to be the e-singlet itself, which is exactly why O survives the projection. Cal #27 held: I steelman $v=0$ below and show precisely why it fails.

The sourced embedding (FK Ch III, spin factor of the symmetric bilinear form — pinned, not reconstructed)

The type-IV Jordan algebra is the spin factor $J = \mathbb{R}e \oplus V_0$, $\dim J = n_{\text{C}} = 5$, so $V_0 = \mathbb{R}^4$, rank 2 (Faraut–Korányi, *Analysis on Symmetric Cones*, Ch. III; the two primitive idempotents $c_{\pm} = \frac{1}{2}(e \pm \hat{u})$, $\hat{u} = v/|v| \in \text{unit sphere of } V_0$ — FK III.1.2, the mechanism K974 verified). - The Jordan identity e is fixed by every Jordan automorphism. $\text{Aut}(J)$ for the spin factor is $O(V_0) = O(4)$, acting trivially on e and as the vector rep on V_0 (FK Ch III — automorphisms of a spin factor are the isometries of the form on V_0 , fixing the unit e). - $\text{Aut}(J) \supseteq \text{SO}(4) = \text{SU}(2)_{\text{L}} \times \text{SU}(2)_{\text{R}}$ (the e-fixing rotations of V_0). So e is $\text{SU}(2)_{\text{L}} \times \text{SU}(2)_{\text{R}}$ -invariant — it spans the (1,1) singlet. - Under this e-fixing $\text{SO}(4) \subset \text{SO}(5)$, the $\text{SO}(5)$ vector $5 = (1,1) \oplus (2,2) [= e \oplus V_0]$, with $V_0 = \mathbb{R}^4$ = the (2,2) ($\text{SO}(4)$ ’s vector rep). This is exactly F603’s $\text{Sp}(1)_{\text{L}} \times \text{Sp}(1)_{\text{R}}$ decomposition ($\text{SO}(5) = \text{Sp}(4) \supset \text{Sp}(1) \times \text{Sp}(1)$); the (1,1) it names IS the e-direction (the vector the e-fixing $\text{SO}(4)$ stabilizes).

The projection — and why the dimension drop does NOT kill O

The spin-factor “vector part” of any $x = \alpha e + v$ is $v = x - (\langle x, e \rangle / \langle e, e \rangle) e$, the orthogonal projection onto $V_0 = e^{\perp}$ (trace form). The $5 \rightarrow 4$ drop removes exactly the e-direction = the (1,1) singlet.

O (F603, target-innocent) = the (2,2) bi-doublet — the Higgs condensate, color-singlet, $\text{SU}(2)_{\text{L}}$ -DOUBLET ** , $Y = \frac{1}{2}$. It lives entirely in the (2,2).

(2,2) and (1,1) are distinct K-irreps $\Rightarrow$ orthogonal under the invariant trace form (Schur). Therefore:

$$\langle O, e \rangle = 0 \Rightarrow \alpha_O = 0 \Rightarrow v_O = \text{proj}_{V_0}(O) = O \neq 0.$$

S2 clears: $v = O \neq 0$, non-central. The dropped dimension is the singlet; O is the bi-doublet; a distinct irrep cannot have a component along the singlet — so O loses nothing in the projection. This is the *sourced* closure of the exact

gap Casey flagged: not “O is generically non-singlet,” but “the dropped direction is the specific irrep (1,1), O is the specific irrep (2,2), and $\langle(2,2),(1,1)\rangle = 0$.”

Steelman $v=0$ (Cal #27 — I do not wave it through)

For $v=0$ we need $O \propto e$, i.e. $O \in (1,1)$ singlet — O central. But O is the ****SU(2)_L doublet**** (the electroweak-symmetry-breaking VEV). A symmetry-breaking condensate is *by definition* not proportional to the identity — if it were, it would break nothing. So $v=0$ would contradict O being the Higgs doublet (F603, target-innocent). $v=0$ is excluded by the physics that pins O’s direction, independent of any mass. (Had O been the (1,1) singlet part of the vector 5 — the piece F603 explicitly does NOT assign to the condensate — then $v=0$ and O would be central. It isn’t.)

Target-innocence (K880 quarantined)

- O’s direction: from quantum numbers alone (color-singlet, SU(2)_L-doublet, $Y=1/2 \rightarrow (2,2)$), F603, independent of the fermions/masses.
- $e = (1,1)$: from the Jordan structure (Aut fixes the identity), FK Ch III.
- The muon mass never enters. The projection is a rep-theory identity on D_{IV}^5 .

Count-node reconciliation (K975 watch-item 1 — resolved by the rank, not the words)

The Jordan frame has 2 primitive idempotents (rank 2); the three generation modes carry support-orbit ranks $\ell = 0$ (tau), 1 (muon), 2 (electron). The correspondence is NOT the inverted “interior/boundary” wording of K947: - Interior idempotent seats (2) = the two nonzero-rank modes = {muon $\ell=1$ (edge/minimal rep), electron $\ell=2$ (full rank)}. - Boundary (+1) = the rank-0 mode = tau ($\ell=0$, singular Shilov measure). - Total = $2 + 1 = 3 = \text{rank}+1$. This supersedes K947’s “interior = {tau, muon}” (that used the inverted Jordan-language). The muon is an interior seat in *both* readings — its promotion is robust, as Keeper flagged. Assign the count node by the rank ℓ ; retire the words.

Handoffs

- @Elie — confirm $v \neq 0$ numerically on the sourced embedding: build O as the (2,2) bi-doublet in the SO(5) vector 5, e as the (1,1) singlet (= the SO(4)-fixed direction), project O onto $V_0 = e^\perp$, verify $|v| = |O| \neq 0$ (and $\langle O, e \rangle = 0$ exactly — the Schur-orthogonality check). This replaces toy 4900’s frame-invariant ($v=0$) symbol and toy 4902’s asserted $v \neq 0$ with the *sourced* computation. Cal’s bar: “is v non-central?” — the answer is yes because $O \not\propto e$ (distinct irreps).
- @Cal — S2’s clearance rests on two sourced facts: (i) e = the (1,1) singlet (Aut(J) fixes the identity, FK Ch III), (ii) O = the (2,2) (F603). The $v \neq 0$ conclusion is Schur orthogonality of distinct K-irreps — please audit that the dropped dimension is genuinely the e-singlet and not a skew embedding (the one way this fails). §121 spin-factor construction is the anchor.
- @Keeper — if Elie’s numeric confirms $v \neq 0$ (it must, by orthogonality), then S1 $\rightarrow$ S3 $\rightarrow$ S4 $\rightarrow$ interior-2 $\rightarrow$ S2 $\rightarrow$ fire K967 blind $\rightarrow$ muon DERIVED. I am NOT banking it — you fire. Count node reconciled to the rank: interior = {muon, electron}, boundary = tau; supersede K947.
- @Casey — the make-or-break lands the good way, and for the reason you’d want: I did not clear it on “O is obviously not a singlet” — you were right that the dimension drop is exactly where that fails. What saves it is sharper — the one direction the projection throws away is *the identity direction itself* (the Jordan e, which is the symmetric singlet), and the Higgs condensate is an SU(2) doublet, a different symmetry-type entirely. Two different symmetry-types are perpendicular, so the condensate has zero overlap with the thrown-away direction — it survives the projection whole. In plain terms: the projection deletes the “average/identity” part, and the Higgs isn’t in the average part, it points sideways. So it selects a frame, the muon sits on one of its two idempotents, and the last gate opens — pending Elie’s number and Cal’s audit, with Keeper firing K967, not me.

Notes only; no toy/theorem claimed (Elie owns the numeric; next_toy=4904 read, untouched). F722: S2 sourced projection. FK Ch III: $J = \mathbb{R}e \oplus V_0$ ($V_0 = \mathbb{R}^4$), Aut(J) fixes $e \Rightarrow e = (1,1)$ singlet; $5 = (1,1) \oplus (2,2)$, $V_0 = (2,2)$. The $5 \rightarrow 4$

drop removes the e-singlet. $O=F603$ (2,2) bi-doublet $\otimes e$ (Schur) $\Rightarrow v=O \neq 0 \Rightarrow S2$ CLEARS non-central, target-innocent. Steelman $v=0$ needs $O \propto e$ (singlet) = contradicts Higgs-doublet; excluded. Count reconciled to rank: interior idempotents={muon $\ell=1$, electron $\ell=2$ }, boundary=tau $\ell=0$, supersedes K947. Elie confirms numeric on sourced embedding; Keeper fires K967 blind $\rightarrow$ muon DERIVED if $v \neq 0$. — Lyra